

	HINDUSTAN PETROLEUM GREEN R&D CENTRE, KIADB INDUSTRIAL AREA, TARABAHALLI VILLAGE, HOSKOTE TALUK, BANGALORE-560067 MATERIAL SAFETY DATA SHEET- HP-COP
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SECTION I - PRODUCT IDENTIFICATION

Trade Name and Synonymous: HP-COP (CO COMBUSTION PROMOTER ADDITIVE)

Chemical Description	Platinum on Aluminium oxide & Cerium oxide
Chemical Formula	Pt/Al ₂ O ₃ /CeO ₂

SECTION II - COMPOSITION/INFORMATION ON INGREDIENTS

INGREDIENT(S)	CAS	Wt % BY WEIGHT
Aluminum Oxide	1344-28-1	90-99
Cerium oxide	1306-38-3	1-10
Platinum	7440-06-4	0.01-0.1

SECTION III - HAZARDS IDENTIFICATION

EMERGENCY OVERVIEW

PHYSICAL APPEARANCE: Light Yellow powder

IMMEDIATE CONCERNS: During a fire, irritating and highly toxic gases may be generated by thermal decomposition or combustion. Causes skin and eye irritation. Prolonged or excessive inhalation may cause respiratory tract irritation. Prolonged or repeated overexposure may cause lung damage.

POTENTIAL HEALTH EFFECTS

EYES	May cause mechanical irritation which can scratch the eye.
SKIN	Prolonged or repeated contact may cause skin irritation
INGESTION	Ingestion of large quantities may cause stomach and intestinal distress.
INHALATION	Inhalation of dust may cause irritation of the nose, throat and lungs. Poison by inhalation of fumes (caused by heating to high temperatures). Fumes will cause inflammation and ulceration of the respiratory tract.

SIGNS AND SYMPTOMS OF OVEREXPOSURE

EYES	Redness, irritation, scratched cornea.
SKIN	Irritation, rash.
INGESTION	Upset stomach, intestinal distress

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INHALATION	Irritation of nose, throat and upper respiratory tract.
CHRONIC EFFECTS	Prolonged or repeated overexposure may cause lung damage.
MEDICAL CONDITIONS AGGRAVATED	Breathing problems, skin problems.
ROUTES OF ENTRY	Inhalation, ingestion, eyes, skin.
TARGET ORGAN STATEMENT:	Respiratory system, lungs, skin, and eyes.

SECTION IV - FIRST AID MEASURES

Symptoms and effects	Dust may be irritating to the respiratory tract and cause symptoms of bronchitis. May be irritating to eyes
First aid	
General	In all cases doubt, or when symptoms persist, seek medical attention.
Inhalation	Move to fresh air, rest half upright position, loosen clothing. Oxygen or artificial respiration if there is difficult in breathing. Always seek medical advice in case of significant exposure.
Skin	Wash immediately with soap and water
Eye	Rinse thoroughly with plenty of water. Eyelids should be held away from eyeball to ensure thorough rinsing. Seek medical advice if irritation Persists
Ingestion	Rinse mouth, give water to drink
Advice to physician	Symptomatic treatment is advised

SECTION V - FIRE- FIGHTING MEASURES

Extinguishing Media	Not applicable. The product is not flammable.
Unsuitable extinguishing media	Not applicable.
Special exposure hazards	Not applicable
Hazardous decomposition/ combustion products	Not applicable.
Protective equipment	Not applicable.

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SECTION VI - ACCIDENTAL RELEASE MEASURES

Personal precautions	Avoid breathing dust. For personal protection see section 8
Environmental precautions	No specific recommendations
Methods for cleaning up	Avoid dust generation. Collect as much as possible in a clean container for (preferable) reuse or disposal. Flush remainder with water.

SECTION VII - HANDLING AND STORAGE

Handling	Areas in which handling generates dust should have adequate ventilation equipment. Minimize exposure to personnel
Fire and explosion prevention	Avoid dust generation. Take precautionary measures against static discharges
Storage requirements	No specific recommendations
Other information	To maintain quantity keep container tightly closed.

SECTION VIII - EXPOSURE CONTROLS/PERSONAL PROTECTION

HAZARDOUS COMPONENTS					
		EXPOSURE LIMITS			
Chemical Name		OSHA PEL		ACGIH TLV	
		ppm	mg/m ³	ppm	mg/m ³
Aluminum Oxide	TWA	[1]	15 mg/m ³ [1]	[2]	1 mg/m ³ [2]
Cerium Oxide	TWA	[1]	15 mg/m ³ [1]	[3]	1 mg/m ³ [3]
Platinum	TWA	[1]	15 mg/m ³ [1]	[2]	1 mg/m ³ [2]

OSHA TABLE COMMENTS:

1. 5 mg/m³ respirable
2. Respirable (as metal, insoluble component)
3. Exposure limits not established

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ENGINEERING CONTROLS:	If user operations generate dust, fume or mist, use ventilation to keep exposure to airborne contaminants below the exposure limit.
PERSONAL PROTECTIVE EQUIPMENT	
EYES AND FACE	Follow facility guidelines in the absence of dusts. Snug-fitting goggles should be worn in dusty work environments.
SKIN	Use of proper hygiene practices in the workplace is recommended.
RESPIRATORY	Wear NIOSH approved particulate filtering respirator rated N, R, or P95 or 100 or equivalent in the absence of proper environmental control. Type of respirator depends on level of exposure.
COMMENTS	All inert or nuisance dusts, whether mineral, inorganic, or organic, not listed specifically by substance name are covered by the Particulates Not Otherwise Regulated (PNOR) limit which is 5 mg/m ³ for respirable fraction and 15 mg/m ³ for total dust. ACGIH exposure guidelines of less than 3 mg/m ³ (respirable) and 10 mg/m ³ (inhalable) have been established for particles (insoluble/poorly soluble) not otherwise specified (PNOS).

SECTION IX - PHYSICAL AND CHEMICAL PROPERTIES

Appearance	Powder
Color	Light Yellow-
Odor	odorless
Boiling Point/ range	>2000 ⁰ C
Melting point/range	>1200 ⁰ C
Density	800 kg/m ³
Solubility in Water	Insoluble

SECTION X - STABILITY AND REACTIVITY

STABLE	Yes, under recommended storage & handling conditions
HAZARDOUS POLYMERIZATION	No
HAZARDOUS DECOMPOSITION PRODUCTS	Not applicable

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SECTION XI - TOXICOLOGICAL INFORMATION

ACUTE

Chemical Name	ORAL LD ₅₀ (rat)
Aluminum Oxide	> 5000 mg/kg (rat)
Cerium Oxide	> 5,000 mg/kg
Platinum	10-110 mg Pt/kg

EYE EFFECTS: rabbit

Result: non irritating

SKIN EFFECTS: rabbit

Result: non irritating

CARCINOGENICITY

Chemical Name	NNTP Status	IARC Status	OSHA Status
Aluminum Oxide	Not listed.	Group 3 - not classifiable	Not listed
Cerium Oxide	Not listed	Not listed	Not listed
Platinum	Not listed	Not listed	Not listed

GENERAL COMMENTS: Not tested in animal expt. The toxicological taken from products of similar composition

SECTION XII - ECOLOGICAL INFORMATION

ECOTOXICOLOGICAL INFORMATION	Aquatic Toxicity (Aluminum Oxide):
96-hr NOEC (<i>Salmo trutta</i>)	> 100 mg/L (OECD Guideline 203)
48-hr NOEC (<i>Daphnia magna</i>)	> 100 mg/L (OECD Guideline 202)
72-hr NOEC (<i>Selenastrum capricornutum</i>)	> 100 mg/L (OECD Guideline 201)
CHEMICAL STATE INFORMATION	This material is of mineral origin. It is not biodegradable

This product should not be used or allowed to enter, drain on water courses.

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SECTION XIII - DISPOSAL CONSIDERATIONS

- DO not allow into drains or water courses
- Contaminated packaging must be emptied of all residues
- Following appropriate cleaning, they must be sent to recycling plant

SECTION XIV - TRANSPORT INFORMATION

DOT (DEPARTMENT OF TRANSPORTATION)

PROPER SHIPPING NAME: Not regulated

ROAD AND RAIL (ADR/RID)

PROPER SHIPPING NAME: Not regulated

AIR (ICAO/IATA)

SHIPPING NAME: Not regulated

VESSEL (IMO/IMDG)

SHIPPING NAME: Not regulated

CANADA TRANSPORT OF DANGEROUS GOODS

SHIPPING NAME: Not regulated

SECTION XV - REGULATORY INFORMATION

UNITED STATES

SARA TITLE III (SUPERFUND AMENDMENTS AND REAUTHORIZATION ACT)

FIRE: No **PRESSURE GENERATING:** No **REACTIVITY:** No **ACUTE:** No **CHRONIC:** Yes

313 REPORTABLE INGREDIENTS: Not Listed.

CERCLA (COMPREHENSIVE RESPONSE, COMPENSATION, AND LIABILITY ACT) CERCLA

REGULATORY: Not Listed.

National Fire Protection Association (U.S.A.):

Health: 2

Flammability: 0

Reactivity: 0

Specific hazard:



Health	2
Fire	0
Reactivity	0
Personal Protection	E

TSCA (TOXIC SUBSTANCE CONTROL ACT)

TSCA STATUS: All components are listed on the TSCA Inventory or are excluded or exempt.

REGULATIONS

STATE REGULATIONS: California

CALIFORNIA PROPOSITION 65: This product does not contain chemical(s) known to the state

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of California to cause cancer, birth defects, or reproductive harm.

RCRA STATUS: This product, if discarded as sold, is not a Federal RCRA hazardous waste. Processing, use or contamination of this product may change the waste management options. State and local disposal regulations may differ from federal disposal regulations.

CANADA

WHMIS HAZARD SYMBOL AND CLASSIFICATION: Does not meet classification criteria pursuant to the Canadian Hazardous Products Act.

WHMIS (WORKPLACE HAZARDOUS MATERIALS INFORMATION SYSTEM): This MSDS has been prepared according to the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all of the information required by the CPR.

Classification according to EEC directives

- Danger Symbol and designation : No danger Symbols
- R-phrases : None
- S-phrases : None

Additional Information

To our present Knowledge the product is not a substance subject to mandatory marking in accordance with EEC-Directive 67/548/EEC or amendments.

SECTION XVI - OTHER INFORMATION

DEPARTMENT issuing data specification sheet: Product safety department

These data are based on our knowledge. However, they shall constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

NON-POISONOUS	: YES
NON-COROSIVE	: YES
NON- INFLAMABLE	: YES
NON-HAZARDOUS	: YES